
Forecast Challenge Desk

Technical Design Review

NorthStar Infrastructure Group
AI Engineer Case Assignment

Cluster	FIN · Finance
Programme	Agentic Operations Programme
Workflow	Monthly Project Forecast Challenge & EAC Review
Status	Prototype — not production deployed
Date	June 2025

Prepared by
Aymane Lamyaghri

All figures derived from mock data.

Contents

1	Finance Workflow: Forecast Challenge Desk	3
1.1	Workflow Selection Rationale	3
1.2	The Problem Being Solved	3
1.3	Future-State Workflow	3
1.4	Prototype Scope and Data Design	4
2	Technical Design Review	5
2.1	Architecture Overview	5
2.2	Agent Architecture — 20 Parallel Independent Agents	5
2.2.1	ReAct Loop — Per-Project Execution	5
2.3	Tool Design — Deterministic by Principle	5
2.4	Signal Computation — Pre-LLM Filtering	6
2.5	Model Selection	7
2.6	Build vs Buy	7
2.7	Evaluation Strategy and Results	7
2.8	Hard Cases	8
2.9	Failure Modes Observed in Testing	8
2.10	Production Infrastructure Stack	8
2.11	RAG and Historical Institutional Memory	9
2.12	Model Strategy — Prompting and Fine-tuning	9
2.12.1	Immediate: Prompt Engineering	9
2.12.2	Medium-term: Fine-tuning (conditional)	10
3	Production Readiness	11
3.1	Data Integration — The Longest Lead-Time Item	11
3.2	Identity, Access Control, and Governance	11
3.3	Reliability and Month-End SLA	11
3.4	Scalability	12
3.5	Cost at Scale	12
3.6	Continuous Evaluation in Production	12
3.7	Multi-Region and Multi-Language	13
4	Executive Summary	14
4.1	The Business Case in Three Numbers	14
4.2	What the System Does and Does Not Do	14
4.3	Costs and ROI	14
4.4	Risks and Mitigations	14
4.5	Rollout Approach	15

5	Assumptions & Kill Test	16
5.1	Assumption 1: EVM Data Is Reliable Enough to Act On	16
5.2	Assumption 2: Agent FLAG Decisions Predict Actual Overruns	16
5.3	Assumption 3: Controllers Will Use the Tool as Designed	16
5.4	Kill Metrics — Hard Thresholds	16
6	Closing Statement	17

1 Finance Workflow: Forecast Challenge Desk

1.1 Workflow Selection Rationale

The selected workflow is the monthly EAC (Estimate at Completion) forecast review, known internally as the *Forecast Challenge Desk*. Each month, Project Managers submit revised cost forecasts to Finance, and a Controller must review every submission, challenge unrealistic forecasts, and escalate the highest-risk projects before month-end close.

Three properties made this the right workflow to target:

- **Volume and time pressure.** With 50–200+ active projects, a Controller spends two to three days per cycle on manual review — a bottleneck that compounds at scale.
- **Materiality of failure.** A missed escalation on a \$40M overrun hits the P&L directly. A single such miss likely exceeds the cost of this programme.
- **Defensible decision taxonomy.** EVM data is quantitative, the decision labels map to real Controller actions, and evidence is always citable in dollar figures rather than qualitative judgement.

1.2 The Problem Being Solved

Without tooling, a Controller reviewing 20 submissions must manually:

- Export EVM data from the project controls system (Primavera P6, SAP PS) — a manual pivot exercise with no audit trail.
- Cross-reference each PM's submitted EAC against system estimates — 20–40 minutes per project.
- Detect non-obvious failures: CPI 1.01 masking SPI 0.82 and \$1.65M in acceleration invoices; project CPI 0.936 concealing a Civil & Structural package at CPI 0.610 that holds 90% of total exposure.
- Audit each change-order register — PMs routinely omit pending variations, \$29.8M of undisclosed exposure across 8 of 20 prototype projects.
- Identify systematic optimism bias across 6 months of submissions, not just the current period.

At a projected 200 projects across 10+ business units and 12 cycles per year, this manual process forces a choice between disproportionate headcount and reduced oversight depth — both carrying unacceptable financial risk.

1.3 Future-State Workflow

The Desk runs as a structured first-pass at the opening of each cycle. The Controller's role shifts from data extraction and pattern detection to decision review and escalation.

- **Ingestion.** EVM snapshot exported from the project controls system; PM narratives and CO updates submitted via structured form.
- **Parallel agent run.** 20 independent agents investigate one project each, using up to 7 specialised tools, and write a FLAG / ACCEPT / NEEDS_EVIDENCE decision with cited evidence.
- **Controller review.** Each decision is reviewed with the full investigation trace visible. Override is always available and requires no justification.
- **Audit trail sealed.** All decisions — agent and Controller-modified — are logged to an immutable record with timestamps, user IDs, tool traces, and evidence strings.

1.4 Prototype Scope and Data Design

The prototype covers the full monthly cycle across a simulated 20-project portfolio. The data is engineered to surface the hard parts of the problem, not to flatter the agent.

- 20 projects across 5 regions, 6 project types, and 3 contract structures (10 lump sum, 7 reimbursable, 3 target cost).
- EVM histories of up to 22 periods with realistic CPI/SPI series, rebaseline events, and S-curve progress — total BAC \$970.3M, system EAC \$1,079.5M.
- 5 hard cases built to defeat threshold classifiers: P07 (CPI 1.010 / SPI 0.820), P13 (CPI 0.945, 97% contingency at 40% complete), P06 (project CPI 0.936 masking Civil & Structural 0.610), P17 (PM EAC \$95M requires future CPI 0.931 from 0.762), P01 (same reassurance phrase across 3 months as CPI fell 0.855 → 0.820).
- Change-order register: 8 projects carry \$29.8M in hidden material COs not in PM EAC — including \$6.3M on P06, \$6.0M on P17, \$3.6M on P01.
- Ground-truth labels for 6 back-test cycles (Jun–Nov 2024), enabling accuracy measurement across the full window.

2 Technical Design Review

2.1 Architecture Overview

The system is a single-tier application with no persistent backend — an intentional prototype constraint. The goal is to demonstrate reasoning quality and decision accuracy, not infrastructure maturity. Each production replacement is identified in Section 3.

Layer	Component	Technology	Role
Presentation	Web UI	Streamlit 1.x	Month selector, run trigger, live decision streaming, investigation trace.
Orchestration	Orchestrator	<code>concurrent.future</code>	Runs the shared scan once; fans out 20 agent threads; aggregates decisions.
Agent	Mini-agent (×20)	Custom ReAct loop	Up to 14 turns, 10 tools, one decision. Isolated per project.
Tools	EVM tools (×10)	Deterministic Python	Mathematics, formatting, register lookups. The LLM never generates a number.
Data	Mock portfolio	In-memory	20 projects × 22 periods × 8 WBS packages + CO register + narratives.
LLM	<code>claude-haiku-4-5</code>	Anthropic API	Classification and evidence synthesis; decides which tools to call and what results mean.

2.2 Agent Architecture — 20 Parallel Independent Agents

The core decision is to run **20 independent single-project agents in parallel** rather than one sequential agent iterating over the portfolio.

Why independent. Each decision depends only on one project’s EVM data, narrative, and CO register — no cross-project reasoning is required. A sequential agent would accumulate context-window overhead linearly with no accuracy benefit, and a single stalled investigation would block the rest.

Why parallel. Wall-clock time falls from the sum of all investigations (~120s sequential) to the deepest single one (~25–40s). With 10 concurrent threads, the bottleneck is the most complex project, not the queue — this is what makes a 30-second cycle possible.

2.2.1 ReAct Loop — Per-Project Execution

- **Shared context injection.** The orchestrator runs `load_all_summaries` and `get_portfolio_exposure_ranking` once and injects both into every agent’s first message, so each agent knows the portfolio risk distribution before investigating.
- **Signal-based routing.** Projects marked `[clean]` ($CPI > 0.97$, $gap < 3\%$, no signals) are accepted immediately. Any `!!` signal enters investigation.
- **Adaptive tool selection.** The agent calls `get_project_detail` first; each later call follows from what the previous one revealed. It calls only the tools warranted.
- **Decision and termination.** The agent calls `write_decision` once, citing specific dollar amounts and CPI values. It is isolated to its assigned project ID; max 14 turns as a hard safety limit.

2.3 Tool Design — Deterministic by Principle

All 10 tools are pure deterministic Python functions: they compute EVM mathematics, format text, and perform register lookups. The LLM produces no numbers — every figure it cites was returned by a tool call, which makes the tool-call sequence the audit trail.

Tool	Trigger	Evidence it provides
load_all_summaries	Shared	CPI, SPI, EAC gap, and signal flags for all 20 projects. Establishes routing.
get_portfolio_exposure_ranking	Shared	Projects sorted by \$ at risk (System EAC – BAC). Proportionality calibration.
get_project_detail	Any !!	Full snapshot: CPI/SPI trend, EAC credibility, implied future CPI, TCPI, per-package WBS CPI, contingency, CO alert.
get_change_order_exposure	CO alert	Full CO register with in-EAC flags; adjusted EAC if hidden COs materialise. Critical on reimbursable contracts.
get_wbs_cost_breakdown	HIDDEN_WBS	Subjected overrun per package. Civil CPI 0.610 → \$13.4M overrun, 90% of exposure on P06.
get_spend_acceleration	SPI_TRAP	Monthly AC trend; acceleration appears 1–2 periods before CPI worsens. Critical for P07.
get_cpi_history	Decline/Step	Month-by-month CPI trajectory; separates structural decline from event-driven drop.
get_pm_eac_history	Low credibility	PM vs system EAC over 6 months with gap % and bias label. Surfaces systematic denial.
get_prior_explanations	Repeated	Three months of narrative side by side with repeated-reassurance detection.
write_decision	Terminates	Commits the decision with evidence; enforces project isolation.

2.4 Signal Computation — Pre-LLM Filtering

Before the LLM touches any project, the scanner computes deterministic signals from EVM mathematics. These drive routing and are the primary mechanism preventing hallucinated risk assessments.

Signal	Condition	What it detects
EARLY_PHASE	< 4 periods or < 20% complete	Insufficient data; routes to NEEDS_EVIDENCE without investigation.
EAC_GAP_LARGE	PM EAC > 10% below system	Optimism gap; forces project detail regardless of CPI.
STRUCTURAL_DECLINE	CPI declining N consecutive periods	Self-reinforcing deterioration. 6+ periods is a near-certain FLAG.
STEP_CHANGE	CPI drop > 0.08 in one period	Discrete event; forces root-cause investigation.
CONTINGENCY_N%	Contingency > 80% consumed	Buffer depletion; critical when % complete is far lower.
SPI_TRAP	CPI > 0.99 and SPI < 0.88	Cost fine, schedule behind; acceleration not yet invoiced.
IMPLIED_FUTURE_CPI	PM needs > 10% CPI recovery	Forecast requires performance never achieved.
CONT_AHEAD	Contingency % > % complete + 20	Buffer consumed faster than progress.
HIDDEN_WBS	Civil CPI < 0.78, project CPI > 0.88	Package disaster masked at project level.
REPEATED_NARRATIVE	Watch phrase in current + prior narratives	PM recycling reassurances as CPI worsens.
REBASELINED	BAC increased mid-project	Prior overrun absorbed; metrics may look artificially clean.

2.5 Model Selection

Selected: `claude-haiku-4-5-20251001`. The task is structured classification with deterministic tool outputs and explicit decision criteria. Haiku’s reasoning is sufficient when the framework is well-specified and tool outputs are structured text. Cost and latency advantages are substantial at any realistic review frequency.

Model	Latency	Cost/run	Suitability	Verdict
<code>claude-haiku-4-5</code>	2–4s	~\$0.08	Sufficient for structured classification	Selected
<code>claude-sonnet-4-6</code>	5–10s	~\$0.80	Marginal gain; 10× cost	Calibration reserve
<code>claude-opus-4-8</code>	12–25s	~\$4.00	Overkill for production	Few-shot only
GPT-4o	5–12s	~\$0.90	Viable; not benchmarked here	Fallback

Model pinning policy

Production pins a specific version string and runs a regression suite against the 20 known cases before accepting any model update. Updates are not breaking, but borderline behaviour (CPI 0.88–0.92) can shift between versions.

2.6 Build vs Buy

Component	Decision	Rationale
LLM inference	Buy (Anthropic)	Delivers the required reasoning. Lock-in is manageable — the tool interface is model-agnostic.
Agent framework	Build (custom)	LangChain/CrewAI obscure tool-call sequences and add dependencies. The custom ReAct loop is under 100 lines and fully inspectable.
EVM mathematics	Build	No off-the-shelf library covers construction EVM at this granularity — core IP.
UI	Buy (Streamlit)	Fastest path to a demo; not production. Replacement: FastAPI + React or Teams tab.
Parallelism	Build (stdlib)	<code>ThreadPoolExecutor</code> suffices for 20 threads. Production uses a managed queue.
RAG, Auth, Storage, Observability	Not built	Out of scope for a static-data prototype; each is specified in Sections 2.11 and 3.

2.7 Evaluation Strategy and Results

Quality is measured against per-cycle ground-truth labels embedded in the data, reflecting expert classification on EVM signals, narrative credibility, contingency, and CO exposure. Six cycles (Jun–Nov 2024) form the back-test window.

Metric	Definition	Target	Kill threshold
Accuracy	Decisions matching ground truth / 20	85%+	< 75% for 3 cycles
False Accept Rate	True flags accepted / total true flags	< 10%	> 15% for 3 cycles
Investigation depth	Avg. tool calls on flagged projects	3–6	< 2 or > 8

Ground-truth distribution (Nov 2024). 8 ESCALATE, 3 CHALLENGE, 7 ACCEPT, 2 NEEDS_EVIDENCE; the 11 action items represent \$109.2M above BAC. A flag-everything classifier scores 55% accuracy at 0% FAR; accept-everything scores 45% at 100% FAR — neither

is useful. The 85% target requires separating acceptable variance from serious risk hidden behind clean headline numbers.

Primary metric: False Accept Rate

Accepting a \$40M overrun is categorically worse than flagging a project that turns out fine. A false flag costs 30 minutes; a false accept costs tens of millions and the credibility of the oversight function. This asymmetry is built into every design decision.

2.8 Hard Cases

Five projects are designed to defeat threshold classifiers — any rule-based system misclassifies at least two.

Project	Scenario	Key evidence the agent must find
P07 Manchester Roads	SPI trap	CPI 1.010 looks on budget, but SPI 0.820 + \$1.65M acceleration invoices + 75% contingency at 52% complete; spend +52% vs prior.
P13 Riyadh Hospital	Contingency lie	CPI 0.945 below most thresholds, but 97% contingency at 40% complete + \$2.3M hidden M&E COs.
P06 Cairo Power	Hidden WBS	Project CPI 0.936; Civil & Structural CPI 0.610 → \$13.4M overrun = 90% of exposure + \$6.3M hidden foundation COs.
P17 Lagos Pipeline	Impossible CPI	PM EAC \$95M requires future CPI 0.931 from 0.762 — not credible. System EAC \$123M, gap \$28M, \$6.0M hidden COs.
P01 North Sea	Repeated narrative	“Recovery expected” across Oct–Dec while CPI fell 0.838 → 0.820; PM EAC \$50.2M vs system \$57.8M.

2.9 Failure Modes Observed in Testing

Each failure mode below motivated a specific tool or prompt addition.

- **SPI trap (P07).** Accepted on ~1 in 6 runs before the acceleration tool existed. *Fix:* `get_spend_acceleration` on `SPI_TRAP`.
- **Borderline CPI (0.88–0.92).** Inconsistent classification depending on tool path. *Fix:* raise `IMPLIED_FUTURE_CPI` whenever a forecast needs >10% improvement.
- **Unquantified hidden WBS.** The agent saw Civil CPI 0.610 but sometimes omitted the dollar figure. *Fix:* `HIDDEN_WBS` as a mandatory trigger for the WBS breakdown.
- **Optimism accepted at face value.** A plausible current narrative masked a persistent EAC gap. *Fix:* `get_pm_eac_history` surfaces the 6-month pattern.
- **Early-phase over-investigation.** *Fix:* `EARLY_PHASE` writes `NEEDS_EVIDENCE` with no tool calls.

2.10 Production Infrastructure Stack

Every in-memory component has a direct production replacement.

Prototype	Production	Operational requirement
In-memory data	ETL → PostgreSQL (versioned)	A June re-run must use June's data; overwriting current state is insufficient.
<code>ThreadPoolExecutor</code>	Celery + Redis / SQS	Retry, dead-letter handling, live queue visibility.
Session state	PostgreSQL append-only	Every decision and override logged with user ID and timestamp. Compliance, not optional.
Streamlit UI	FastAPI + React / Teams	Project-scoped Controller views, read-only PM access, leadership summary only.
<code>os.getenv()</code> key	Key Vault / Secrets Manager	Rotation, access audit, least-privilege; no plaintext credentials anywhere.
<code>print()</code>	Langfuse / Lang-Smith	Per-run token cost, latency, prompt hash, error rate.
No auth	Azure AD SSO + RBAC	Hard requirement before any real-data use.
No output check	Tool-use schema validation	Validate every FLAG comment contains a dollar amount and CPI value.

2.11 RAG and Historical Institutional Memory

Every prototype decision is made in isolation — acceptable on static data, a real limitation in production.

- **Historical decision retrieval.** Retrieve the 3 most similar past cases by EVM profile; if all overran after showing CPI 0.88 in month 11, that strengthens a FLAG.
- **PM performance index.** Index each PM's full submission history — EAC accuracy, optimism frequency, CO disclosure rate.
- **False-narrative library.** Index narratives that preceded confirmed overruns; flag semantic matches, replacing the keyword-only check.
- **Implementation.** `pgvector` is most pragmatic — PostgreSQL is already needed for the audit trail, avoiding a separate component.

When RAG becomes necessary

After 6–12 months, NIG will have 1,000+ labelled examples and accumulated overrides — real signal sitting in a database the agent never consults. RAG is a month-six requirement, not a day-one one.

2.12 Model Strategy — Prompting and Fine-tuning

2.12.1 Immediate: Prompt Engineering

- **Few-shot examples.** Embed one clear FLAG (with figures), one clean ACCEPT, one NEEDS_EVIDENCE. The highest-leverage fix for the borderline CPI inconsistency.
- **Chain-of-thought before decision.** Require an explicit “EVM signal: X. CO exposure: Y. PM credibility: Z. Therefore:” line, which also eases Controller review.
- **Critic agent for borderline cases.** Where a comment hedges, route to a second lightweight agent to challenge it — borderline cases only, to control cost.

2.12.2 Medium-term: Fine-tuning (conditional)

Viable once NIG has 2,000–5,000 labelled examples.

- **What to tune:** NIG-specific patterns — contract-type weighting, regional EVM variation, CO terminology. The base model still handles tool selection.
- **When not to:** an unbalanced dataset overfits the majority class. Below 1,000 examples, few-shot prompting wins.
- **Alternative:** DSPy-style prompt optimisation closes 60–70% of the gap to a fine-tuned model with no labelled data. Evaluate first.

3 Production Readiness

The gap to production is real and well-defined — and it is not primarily a model problem. The model works. The gap is data access, identity, reliability under deadline, cost governance, and adoption.

3.1 Data Integration — The Longest Lead-Time Item

Data integration is the most expensive and highest-risk item on the path. Every signal derives from EVM data; if that data is stale or inconsistently coded, the signals are wrong with numerical precision — more dangerous than obviously wrong output.

- **EVM source.** Primavera P6, SAP PS, or equivalent. A monthly ETL extracts AC, PV, EV, and BAC per project and *versions* period snapshots. *6–10 weeks*.
- **PM submission channel.** Structured form with required schema fields. *2–4 weeks*.
- **Change-order register.** Live integration with the contract system (Unifier, Aconex) refreshed daily. *4–8 weeks*.
- **Historical retention.** 6-month backfill if the warehouse keeps only current-period state. *2–3 weeks*.
- **Data quality audit.** Compare system EAC against actual final cost for projects closed in 24 months; if MAE > 15%, remediation must precede deployment.

Risk

EVM data quality is the single biggest production risk. Inconsistent coding across regions produces results Controllers cannot trust. The Phase 1 audit is not optional.

3.2 Identity, Access Control, and Governance

PM and EVM data are commercially sensitive. Access control must precede any real-data use.

- **Authentication.** Azure AD / SSO, no local credentials. The IdP role claim sets portfolio scope; anonymous access blocked at the infrastructure level.
- **Roles.** Controller (full read + override), Leadership (summary only), PM (read-only own decision after review), Administrator (config + audit log, no project data).
- **Key management.** API key in Key Vault / Secrets Manager, 90-day rotation, all access logged.
- **Data residency.** GDPR (EU), Middle East sovereignty, and APAC variation require legal review before each regional rollout.
- **Audit trail.** Immutable append-only store with UTC timestamp, user ID, action, project ID, decision, and evidence string.

3.3 Reliability and Month-End SLA

The review has a hard deadline. Silent failure on committee day forces emergency manual review under exactly the time pressure most likely to produce missed flags.

- **API resilience.** Exponential backoff with jitter; circuit breaker (open after 3 failures/60s, half-open after 2 min); 429 pauses the pool, never terminates the run. Max 4 retries, 45s timeout.
- **Graceful degradation.** If the LLM is down: finish tool-only pre-computation, mark all projects MANUAL_REVIEW_REQUIRED, notify with the signal summary.

- **Run idempotency.** A re-run creates a new run ID, never overwriting the prior run; the audit trail stays immutable.
- **Monitoring.** Alert on any run >3 min, any cycle with FAR >10%, any flagged project with zero tool calls, any Controller below 5% override for two months.
- **Runbook.** Cover re-running a failed cycle, manual override, prompt rollback, and triggering the data audit. Tested before Phase 3.

3.4 Scalability

The current architecture scales to ~50 projects before `ThreadPoolExecutor` hits contention and rate-limit saturation. 200+ requires a distributed queue.

Component	Prototype (20)	Phase 2 (50)	Phase 3+ (200+)
Job dispatch	ThreadPoolExecutor	Celery + Redis	Celery + SQS, auto-scaling
Data layer	In-memory dict	PostgreSQL single	PostgreSQL replicas + pooling
Storage	Session state	Run-history table	+ S3/Blob archival
UI	Streamlit	FastAPI + React	Multi-tenant, region-aware
Observability	Print	Langfuse basic	Enterprise + Grafana
Deployment	Local process	Docker, 1 region	Kubernetes, multi-region
Auth	None	Azure AD, 1 tenant	Multi-tenant RBAC
Run time	25–40s	35–60s	60–120s (rate-limit bound)

Rate limits are the primary constraint at 200+: 200 simultaneous agents need a dedicated tier or capped concurrency (e.g. 30 workers in batches, 6–8 minutes). Acceptable monthly; not for daily monitoring.

3.5 Cost at Scale

Scenario	Proj.	Runs	LLM/run	Annual	Cost driver
Prototype	20	12	~\$0.08	~\$1	Engineering time
Pilot (1 BU)	50	12	~\$0.20	~\$2	Data integration
FIN rollout	200	12	~\$0.80	~\$10	Maintenance
NIG-wide	500	12	~\$2.00	~\$24	Multi-region infra
Daily (FIN)	200	260	~\$0.80	~\$208	Rate-limit tier

Cost reality

LLM API cost is negligible at every realistic scale. The true cost is engineering: data integration (\$150–250K), auth/RBAC (\$75–100K), audit/compliance (\$50–75K), and maintenance (0.5 FTE/year). Total: \$300–450K one-time; \$80–120K/year.

3.6 Continuous Evaluation in Production

A classifier without outcome feedback drifts silently. This pipeline runs from day one.

- **Six-month-lag ground truth.** Each Month-N ACCEPT is re-checked at N+6 against actual cost; an overrun >5% of BAC relabels it a missed FLAG.
- **Regression gate.** Before any prompt change, run the 20 prototype cases plus the accumulated set; block deployment if accuracy < 83% or FAR > 12%.

- **Override signal.** Tag each override false FLAG or missed FLAG; monthly aggregation surfaces systematic bias for prompt calibration.
- **Drift detection.** Track the decision distribution month-over-month; a sudden shift without a health change signals prompt or pipeline drift.

3.7 Multi-Region and Multi-Language

- **Narrative language.** Middle East (Arabic), West Africa (French), APAC (Mandarin/Bahasa). *Recommendation:* translation pre-processing for Phase 3 with language-specific benchmarking.
- **EVM standards.** The WBS tool assumes 8 packages; production needs a mapping layer to normalise regional conventions.
- **Regulatory variation.** Some jurisdictions require AI-assistance disclosure; the audit trail and Controller sign-off satisfy human-in-the-loop in most frameworks.
- **Data residency.** GDPR/PDPA and local rules may restrict processing location. Options: EU endpoint, on-premise deployment, or anonymisation. Legal review precedes each rollout.

4 Executive Summary

The Forecast Challenge Desk reviews NIG’s monthly cost forecasts in under 40 seconds, surfacing the highest-risk projects with cited evidence before a Controller begins their review. It cuts review time from 2–3 days to 2–3 hours per cycle. On the 20-project portfolio (\$970M BAC, \$109M exposure above budget), it identifies all 11 action items, including 5 hard cases that defeat any threshold classifier. LLM cost is under \$10/year at full scale; the real investment is data integration — two engineers, two quarters. The system does not replace Controllers; it gives them a transparent, overridable, auditable starting point.

4.1 The Business Case in Three Numbers

- **\$109M — portfolio exposure above BAC.** The stake the review protects. Missing two high-risk projects is a potential eight-figure P&L hit.
- **240 days — Controller time recovered per year** (10 Controllers × 12 cycles). \$180–240K in capacity redirected to judgement-intensive work.
- **< \$10/yr — LLM cost at full scale.** The constraint is data-integration engineering, not API cost.

4.2 What the System Does and Does Not Do

- **Does:** investigate every project with the same depth every month. **Does not:** make binding decisions — every output requires sign-off.
- **Does:** surface hard-to-find evidence — acceleration before CPI worsens, hidden WBS disasters, repeated narratives, undisclosed COs. **Does not:** verify PM claims or access data outside the EVM and CO inputs.
- **Does:** cite specific figures in every comment. **Does not:** replace the Controller’s project knowledge or escalation authority.
- **Does:** provide a documented audit trail. **Does not:** improve outcomes if EVM data is stale or inconsistently coded.

4.3 Costs and ROI

Category	Estimate	Notes
LLM API (full scale)	< \$10/year	claude-haiku-4-5, 200 projects × 12 cycles.
Initial engineering	\$300–450K	Data integration, auth/RBAC, audit/compliance — 2 engineers × 2 quarters.
Maintenance	\$80–120K/year	0.5 FTE for prompt governance, model updates, pipeline monitoring.
Controller time recovered	240 days/year	At £800–1,000/day: £192–240K/year equivalent.
Risk avoided	Hard to quantify	One missed \$40M escalation likely exceeds the full programme cost.

4.4 Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
EVM data quality insufficient	Medium	High	Data audit before Phase 2; proceed only if MAE < 15%.
PMs game the system	Medium	Medium	Transparent evidence; CO withholding addressed by direct contract-system integration.
Controllers stop reviewing critically	Medium	High	Monitor override rate; mandate rationale for ACCEPT below CPI 0.92.
Model drift after update	Low	Medium	Pin version; regression suite gates deployment.
Data-residency non-compliance	Low	High	Legal review per region; EU endpoint or on-premise.

4.5 Rollout Approach

- **Phase 1 — Parallel pilot (M1–2).** 1 Controller, 20 projects, mock or anonymised data. *Gate:* FAR < 15%, override 10–60%, positive assessment.
- **Phase 2 — Controlled expansion (M3–4).** 5 Controllers, 50 projects, real EVM integration, audit trail live. *Gate:* FAR < 12%, override stable, data within 3 days of close.
- **Phase 3 — Full FIN rollout (M5–8).** All Controllers, full portfolio, multi-region. *In production* — monitor FAR, override, run time, contest rate.
- **Phase 4 — Cross-cluster (M9–12).** Extend to budget variance, cash flow, earned schedule. *Gate:* Phase 3 FAR below 10%, adoption above 80%.

5 Assumptions & Kill Test

The programme depends on defining kill criteria before requesting further investment. Three load-bearing assumptions follow; if any is false, the solution cannot work as designed or should not be trusted with real decisions.

5.1 Assumption 1: EVM Data Is Reliable Enough to Act On

Why it matters. Every signal derives mathematically from EVM data. If that data is stale, miscoded, or manually adjusted without trail, the agent applies rigorous reasoning to bad inputs — producing *precisely* wrong output delivered with numerical confidence.

How to validate. Before Phase 2, reconstruct EVM data at each monthly close for 24 months and compare system EAC against actual final cost on completed projects. If MAE exceeds 15% of BAC, quality is insufficient and remediation must precede Phase 2.

Consequence if false. The quantitative layer must fall back to qualitative signals — a lower-precision, higher-FAR system — and NIG's EVM function needs remediation independent of this initiative.

5.2 Assumption 2: Agent FLAG Decisions Predict Actual Overruns

Why it matters. The prototype is scored against expert labels, not actual outcomes. The real test is whether flagged projects overrun more than accepted ones.

How to validate. A six-month-lag study from Month 1: at Month 7, compare actual cost-to-date against BAC and PM EAC. FLAG projects should overrun at $\geq 3\times$ the rate of ACCEPT. Below $2\times$, the agent adds no predictive value.

Consequence if false. The agent reasons correctly about signals that do not predict outcomes — a systemic EVM-discipline problem requiring a programme-level intervention, not an agent fix. The agent should be suspended until discipline is established.

5.3 Assumption 3: Controllers Will Use the Tool as Designed

Why it matters. Value depends on critical review. *Over-reliance* (approving without reading) recreates a single point of failure; *under-reliance* (dismissing every flag and reviewing manually anyway) adds process without value.

How to validate. Track override rate per Controller. Healthy: 10–40%. Below 10% is over-reliance; above 60% is trust failure. Add structured interviews at months 1, 3, and 6.

Consequence if false. Over-reliance: mandate written rationale for ACCEPT below CPI 0.92. Under-reliance: redesign the comment to lead with the single most important finding and add a “what changed since last month” view.

5.4 Kill Metrics — Hard Thresholds

These define when the solution must be suspended or redesigned — with specific windows and actions.

Kill Metric	Threshold	Window	Action
False Accept Rate	> 15%	3 cycles	Suspend pending root-cause analysis; resume only when FAR < 10% in testing.
Override Rate	< 5% or > 60%	2 months, any Controller	Below 5%: review sessions, written rationale. Above 60%: interview, evidence-format redesign.
EVM Data Lag	> 2 weeks stale	Any run	Run blocked; all projects MANUAL REVIEW REQUIRED. Data SLA contractually enforced.

6 Closing Statement

The Forecast Challenge Desk is not a speculative AI demo; it is a practical control mechanism for a high-value finance workflow where speed, evidence, and auditability matter. The prototype shows that an agentic system can review a portfolio consistently, identify hidden forecast risk, and produce Controller-ready evidence without making binding decisions.

The strongest argument for the system is not that it replaces human judgement, but that it protects it. Controllers should spend their time challenging assumptions, validating evidence, and escalating material risk — not manually reconstructing CPI trends, hidden change-order exposure, or repeated optimism patterns across dozens of projects.

The path to production is clear: integrate reliable EVM data, enforce identity and audit controls, monitor false accepts aggressively, and keep the Controller firmly in the loop. If those conditions are met, the Desk becomes a scalable first-pass review layer that improves forecast discipline, reduces month-end pressure, and lowers the probability of costly missed escalations.